

```
In [1]: import pandas as pd
```

```
In [2]: data = {
    'Variable': [
        'Climate',
        'Housing',
        'Health',
        'Crime',
        'Transportation',
        'Education',
        'Arts',
        'Recreation',
        'Economy'
    ],
    'PC1': [
        0.190,
        0.544,
        0.782,
        0.365,
        0.585,
        0.394,
        0.985,
        0.520,
        0.142
    ],
    'PC2': [
        0.017,
        0.020,
        -0.605,
        0.294,
        0.085,
        -0.273,
        0.126,
        0.402,
        0.150
    ],
    'PC3': [
        0.207,
        0.204,
        0.144,
        0.585,
        0.234,
        0.027,
        -0.111,
        0.519,
        0.239
    ]
}

df = pd.DataFrame(data)

print(df)
```

	Variable	PC1	PC2	PC3
0	Climate	0.190	0.017	0.207
1	Housing	0.544	0.020	0.204
2	Health	0.782	-0.605	0.144
3	Crime	0.365	0.294	0.585
4	Transportation	0.585	0.085	0.234
5	Education	0.394	-0.273	0.027
6	Arts	0.985	0.126	-0.111
7	Recreation	0.520	0.402	0.519
8	Economy	0.142	0.150	0.239

```
In [3]: loading_values = df.set_index('Variable').abs()

print("Absolute Loading Values:")
print(loading_values)
```

Absolute Loading Values:

	PC1	PC2	PC3
Variable			
Climate	0.190	0.017	0.207
Housing	0.544	0.020	0.204
Health	0.782	0.605	0.144
Crime	0.365	0.294	0.585
Transportation	0.585	0.085	0.234
Education	0.394	0.273	0.027
Arts	0.985	0.126	0.111
Recreation	0.520	0.402	0.519
Economy	0.142	0.150	0.239

```
In [4]: pc1_variable = loading_values['PC1'].idxmax()
pc2_variable = loading_values['PC2'].idxmax()
pc3_variable = loading_values['PC3'].idxmax()

print("Variable contributing most to PC1:", pc1_variable)
print("Variable contributing most to PC2:", pc2_variable)
print("Variable contributing most to PC3:", pc3_variable)
```

Variable contributing most to PC1: Arts
 Variable contributing most to PC2: Health
 Variable contributing most to PC3: Crime

```
In [5]: print("Principal Component 1:")
print(pc1_variable, "=", df.loc[df['Variable'] == pc1_variable, 'PC1'].iloc[0])

print("\nPrincipal Component 2:")
print(pc2_variable, "=", df.loc[df['Variable'] == pc2_variable, 'PC2'].iloc[0])

print("\nPrincipal Component 3:")
print(pc3_variable, "=", df.loc[df['Variable'] == pc3_variable, 'PC3'].iloc[0])
```

Principal Component 1:
 Arts = 0.985

Principal Component 2:
 Health = -0.605

Principal Component 3:
 Crime = 0.585